



HAMDARD UNIVERSITY BANGLADESH

Department of Computer Science and Engineering
Faculty of Science, Engineering and Technology

Course Code: MAT317

Course Title: Engineering Mathematics

Final Examination

Semester: Fall 2025

Date: 3/5/2026

Total Time: 2:00 Hours

Total Marks: 40

Answer any four (04) out of the following questions:

4 × 10 = 40

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

No.	Questions	CLOs	Marks
Q1.	(a) Define finite Fourier sine and cosine transform.	CLO1	[2]
	(b) Find the Fourier sine transform of the following functions:	CLO3	[4]
	$F(x) = \begin{cases} x; & 0 < x < 1 \\ 2 - x; & 1 < x < 2 \\ 0; & x > 2 \end{cases}$		
	(c) Find the finite Fourier cosine transform of the following functions:	CLO3	[4]
	$F(x) = 2x, 0 < x < 6$		
Q2.	(a) Find the Fourier transform of the following functions:	CLO3	[6]
	$F(x) = \begin{cases} 1; & x < a \\ 0; & x > a \end{cases}$		
	Hence evaluate $\int_{-\infty}^{\infty} \frac{\sin pa \cos px}{p} dp$ and $\int_0^{\infty} \frac{\sin p}{p} dp$.		
	(b) Find the Fourier integral of the function $f(x) = e^{-kx}$ and $x > 0$ and $f(-x) = -f(x)$ for $k > 0$ and hence prove that,	CLO3	[4]
	$\int_0^{\infty} \frac{u \sin ux}{k^2 + u^2} du = \frac{\pi}{2} e^{-kx}.$		
Q3.	(a) Define Legendre's polynomial of degree n .	CLO1	[1]
	(b) State and prove Rodrigue's formula for Legendre polynomial.	CLO2	[6]
	(c) Using generating function of Legendre's polynomial prove that		[3]
	$P_n(1) = 1 \text{ and } P_n(-1) = (-1)^n.$		
Q4.	(a) Solve the power series solution:	CLO3	[4]
	$(1 - x^2)y'' - 2xy' + 2y = 0.$		
	(b) Solve the series solution:	CLO3	[6]
	$x^2y'' + xy' + (x^2 - n^2)y = 0.$		
Q5.	(a) Define Bessel function of the first kind of order n .	CLO1	[1]
	(b) Establish the generating function for Bessel function.	CLO2	[5]
	(c) Prove that:	CLO3	[4]
	$J_{n-1}(x) - J_{n+1}(x) = 2 \frac{d}{dx} J_n(x) \text{ and } J_{n-1}(x) + J_{n+1}(x) = 2 \frac{d}{dx} J_n(x).$		
Q6.	(a) State and prove generating function for Legendre's polynomial.	CLO2	[4]
	(b) Express $f(x) = x^4 + 2x^3 + 2x^2 - x - 3$ in term of Legendre's polynomial.	CLO3	[3]
	(c) If $f(x) = \begin{cases} 0; & -1 < x < 0 \\ 1; & 0 < x < 1 \end{cases}$ then prove that,	CLO3	[3]
	$f(x) = \frac{1}{2} P_0(x) + \frac{3}{4} P_1(x) - \frac{7}{16} P_3(x) + \dots$		

- | | | | |
|------|---|-------|---|
| (b). | Write the relationship between intra-domain routing and inter-domain routing algorithm. | CLO 5 | 4 |
| 7. | (a). Define address Resolution Protocol (ARP). Write the packet format of ARP protocol. | CLO 5 | 4 |
| | (b). What is Hop count? Explain all the types of Routing Information Protocol timers. | CLO 5 | 4 |



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 317: Computer Networks

Batch: 25th

Semester: Fall 2025

Time: 2 Hours

Date: 08/03/2026

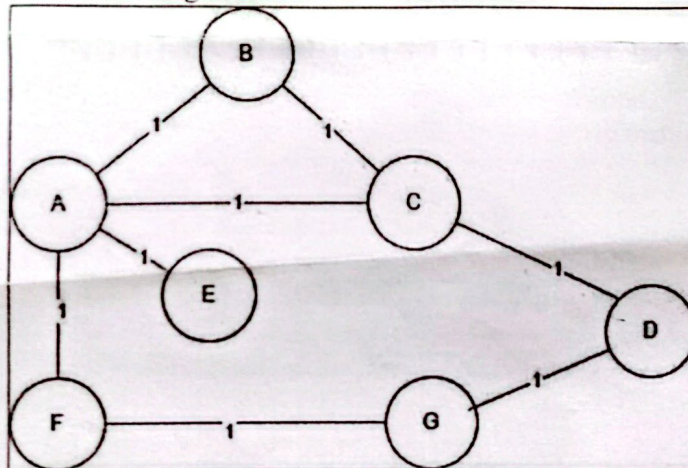
Final Examination

Total Marks: 40

Answer any **five** questions of the following.

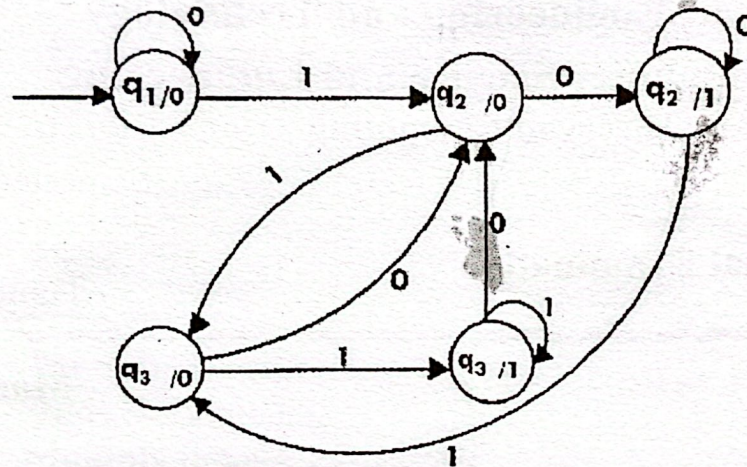
5 × 8 = 40

- | | CLOs | Marks |
|--|-------|-------|
| 1. (a). Briefly explain the relay agent of Dynamic Host Configuration Protocol (DHCP). | CLO 4 | 2 |
| (b). Explain the working procedure of DHCP protocol. | CLO 5 | 6 |
| 2. (a). What is Domain Name System (DNS) in Computer Networks? | CLO 3 | 2 |
| ✓ (b). Explain the steps to find the DNS to IP address or vice-versa. | CLO 3 | 6 |
| 3. (a). Define all types of addresses of Network Address Translation (NAT) protocol. | CLO 4 | 2 |
| ✓ (b). Explain types of NAT's with pros and cons. | CLO 4 | 6 |
| 4. (a). Define single and multi-area Open Shortest Path Fast (OSPF). | CLO 5 | 2 |
| ✓ (b). Consider the following network: | CLO 5 | 6 |



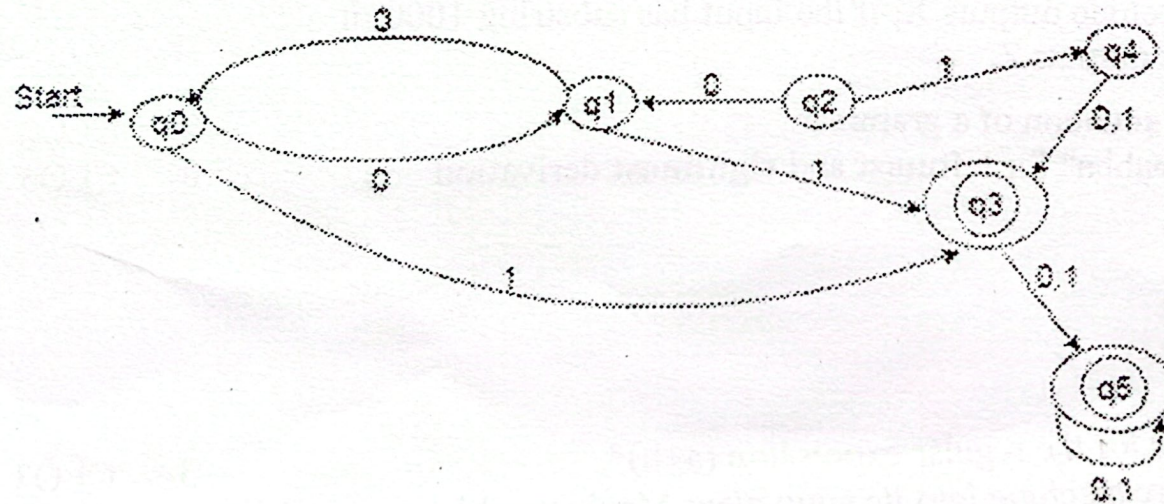
Calculate the shortest path using link state routing protocol.

- | | | |
|--|-------|---|
| 5. (a). What is Socket Programming in Computer Networks? | CLO 3 | 2 |
| (b). Explain all the error-reporting messages of Internet Control Message Protocol (ICMP). | CLO 4 | 6 |
| ✓ (a). Write the differences between distance vector routing protocol and link state routing protocol. | CLO 5 | 4 |



7. Using DFA minimization techniques, minimize state(s) of the given DFA.

8 CLO3





Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE311: Theory of Computing

Batch: 25th

Semester: Fall 2025

Date: 01/03/2026

Final Examination

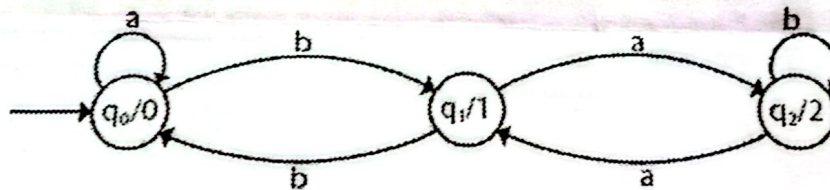
Time: 2 Hours

Total Marks: 40

Answer any five questions.

Marks: 5×8 = 40

- | | Marks | CLOs |
|--|-------|------|
| 1. (a) Define the Mealy machine, mentioning its possible tuples. | 2 | CLO2 |
| (b) Design a Mealy machine for a binary input sequence such that if it has a substring 1111, the machine outputs X; if the input has substring 1000, it outputs Y; otherwise it outputs Z. | 6 | CLO3 |
| 2. (a) Define the ambiguous situation of a grammar. | 2 | CLO2 |
| (b) Derive the string "aabbabba" for leftmost and rightmost derivation using a CFG given by,
$S \rightarrow aB \mid bA$
$A \rightarrow a \mid aS \mid bAA$
$B \rightarrow b \mid bS \mid aBB$ | 6 | CLO3 |
| 3. (a) Derive production rules for the regular expression $(a+b)^*$. | 3 | CLO3 |
| (b) Convert the given Moore machine into its equivalent Mealy machine: | 5 | CLO3 |



- | | | |
|---|-------|------|
| 4. Suppose,
$S \rightarrow (L) \mid a$
$L \rightarrow L, S \mid b \mid s$
Find leftmost derivation for
(i) $(a, (a, a))$
(ii) $(a, ((a, a), s))$ | 4×2=8 | CLO3 |
| 5. Design a Pushdown Automata (PDA) for accepting a language wCw^R , where $w = \{a, b\}$. Show that "baaCaab" is accepted by the developed PDA. | 8 | CLO3 |
| 6. Convert the following Mealy machine into an equivalent Moore machine: | 8 | CLO3 |

Moore

Mealy

- i) Using SQL, create a table Std_details (std_id, std_name, std_email, std_contact, std_bg, ob_mark) from the Student table.
- ii) Derive the SQL to insert Siam's std dob = '05-05-1995' who is in EEE in the table std_details.
- iii) Using the table Student, Write the SQL to create a view Std_vu (std_id, std_name, std_email, std_contact, std_addr, std_bg) for CSE department.

6. (a) Briefly explain the different parts of SQL.

CLO 2 2

(b) Write the SQL and Relational Algebra to determine different types of outer-joins for the following relations:

CLO 4 6

People			
Name	Age	meal	
Bill	22	Hamburger	
Sally	28	Pizza	
Marry	23	Soft-drinks	
Tony	32	Shrimp	

Menu	
meal	Day
Hamburger	Monday
Pizza	Tuesday
Chicken	Wednesday
Pasta	Thursday
Tacos	Friday

7. Consider the following **Order** table with variable repeating groups. Find 1NF, 2NF and 3NF for the following order table:

CLO 4 8

Order								
Order id	Order date	Customer	Customer email	Grade	Credit points	Item name	Qty reqd	Unit price
1	11/3/98	Joe Smith	joe@iewr.com	A	1	Hammer	5	\$25
1	11/3/98	Joe Smith	joe@iewr.com	A	1	Saw	10	\$30
1	11/3/98	Joe Smith	joe@iewr.com	A	1	Nails	15	\$40
2	12/3/98	Sam	sam@ter.com	B	4	Saw	20	\$30
2	12/3/98	Sam	sam@ter.com	B	4	Nails	25	\$40



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology Department of Computer Science and Engineering CSE 313: Database Management System

Batch: 25th

Date: 10/03/2026

Final Examination

Semester: Fall 2025

Time: 2 hours

Total Marks: 40

Answer any five questions from the followings.

Marks: 5 x 8 = 40

1. (a) Write the SQL to create a sequence named **dept_deptid_seq** which starts with 100, increments by 10 and the max value is 1000. CLO 3 2
(b) Draw the syntax to insert and modify a sequence in DBMS. CLO 3 3
(c) What is meant by a sequence in DBMS? Write the guidelines for modifying a sequence. CLO 1 3
2. (a) Define the three tier architecture of a Distributed Database Management System. CLO 1 3
(b) State the advantages of Distributed Database Management System. CLO 1 5
3. (a) Let's assume that we have a table named **Employee** with four columns namely **Employee_Name, Designation, Branch and Salary**. CLO 4 3
Now, Write the queries to find the followings:
i) Lowest Salary among all branches where the value of Designation is Asst. Manager.
ii) Highest Salary regardless of Designation.
iii) Number of Employees where Branch name is Uttara.
(b) Define relational algebra. Considering R and S as two relations. Now, write the SQL to find the set theory operations for the given relations R and S. CLO 4 5

R:

First	Last	Age
Bill	Smith	22
Sally	Green	28
Mary	Keen	23
Tony	Jones	32

S:

First	Last	Age
Forest	Gump	36
Sally	Green	28
Donja	De Marco	27

4. Consider the following Employees table:

Emp_id	First_name	Last_name	Email	Job_id	Salary	Department_id
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- (a) By using the Employees table, Write the SQL to show the last name of all employees whose salary is greater than Abel's using a subquery. CLO 4 3
- (b) By using the Employees table, Write the SQL to show the last_name, job_id and salary whose job_id is the same as employee_id 141 and whose salary greater than employee_id 143's salary. CLO 4 5

5. (a) Explain the components of an E-R diagram. CLO 1 2

- (b) Consider the following Student table: CLO 4 6

Std_id	Std_name	Std_email	Std_contact	Std_addr	Std_bg	Std_dept	ob_mrk
3151003	Afia Khan	a@gmail.com	01913000000	Dhaka	A +ve	CSE	72
3151004	Rabeya	b@yahoo.com	01721111111	Sonargaon	O +ve	CSE	66
5551001	Siam	c@hub.eu.bd	01821222222	Dinajpur	B +ve	EEE	56
5362004	Sadia	d@gmail.com	01911333333	Narayanganj	O +ve	Econ	52
2221003	Siam	e@hub.eu.bd	01551444444	Munsigonj	O -ve	BBA	56



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 315: Microprocessor & Embedded System

Batch: 25th

Semester: Fall 2025

Date: 27/02/2026

Final Examination

Time: 2 Hour

Total Marks: 40

Answer any five questions of the following.

5 × 8 = 40

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|----|-----|--|------|---|
| 1. | (a) | What is an Embedded System? Explain the components of Embedded System | CLO3 | 4 |
| | (b) | Draw the hardware architecture of Embedded System. Explain the classifications of Embedded System | CLO3 | 4 |
| 2. | (a) | Mention Four differences between Microprocessor and Microcontroller. | CLO3 | 4 |
| | (b) | Draw the architecture of Embedded System Hardware for Home Control Automation like Smart Home and briefly explain the key parts. | CLO3 | 4 |
| 3. | (a) | Explain the followings pin of Microprocessor 8086:
i. Address Status Line
ii. NMI
iii. Status Signals
iv. INTR | CLO1 | 4 |
| | (b) | Explain Processor Technology and its three types with figures. | CLO3 | 4 |
| 4. | (a) | Mention Types of Interrupt in 8086. | CLO2 | 2 |
| | (b) | Explain Interrupt Execution with the help of Interrupt vector table in the 8086 microprocessors. | CLO3 | 3 |
| | (c) | Briefly Mention Hardware Interrupt and Interrupt Priority in 8086. | | 3 |
| 5. | (a) | What are Actuators? Explain how actuators are natural complements to sensors | CLO3 | 4 |
| | (b) | Briefly explain the following sensors type with example:
i. Biosensors
ii. Occupancy and Motion
iii. Flow
iv. Chemical | CLO3 | 4 |
| 6. | (a) | Draw the diagram for a Micro Program Control Unit and Explain the Instruction Cycle for [MOV X ₁ , X ₂] | CLO1 | 4 |
| | (b) | Analyze Wilkes Design for a Micro Programmed CU for (001) and (011) input. | CLO3 | 4 |